Health at Home

QR-based family health app (phone-first, dark theme)

Project brief for GitHub | Intercollege competition | Version 2.0 | 9 November 2026

Item	Detail
Goal	Rebuild the reference screen recording as a real web app: a family home screen with 5 colour-coded member cards, and a health card for each person. A QR code opens the app. Full records unlock with an access code.
Family	The Sharma family, Delhi (fictional): Aarav (5), Ananya (15), Meera (mother), Rohan (father), Kamla (grandmother, 72). Health IDs MQ-001 to MQ-005.
Hosting	GitHub Pages (free, static). No server, no database, no real patient data.
Encryption	Full records are AES-256-GCM encrypted. Key derived from the access code with PBKDF2-SHA256 (600,000 iterations), using the browser's Web Crypto API.
Stack	HTML, CSS, vanilla JavaScript. A small Node.js script encrypts the data and creates the QR image.

Important - read first

- **Every record here is invented.** Names, IDs, phone numbers, doctors and test values are fictional. Use them for the competition.
- **Never put a real family's medical data in a public GitHub repository**, encrypted or not. Anyone can download an encrypted file and try codes offline (see Security).
- This is a demo and not medical advice. In an emergency the app tells the user to call 108.

1. Reference design (from the screen recording)

The recording is a 40-second phone screen capture of a dark-themed app titled 'Health at Home'. This section lists what it shows so the developer can match it. Colours are approximate, taken by eye from the video.

Home screen

- Large teal hero card: house emoji, title 'Health at Home', text 'Tap a family member to view their health card and update it.'
- Two-column grid of member cards, each in its own colour, with an emoji avatar, name, 'age - sex', and a group label (Paediatrics, Adolescent, Adult - Mother, Adult - Father, Elder).
- Footer note: 'Demo records with fictional data. Changes are saved on this phone. In an emergency call 108.'

Element	Approx. colour	Used for
Page background	#0F1720 (near-black)	Whole app
Panel / card surface	#162029	Sections on member page
Aarav	#0E86A3 teal-blue	Card, header, buttons
Ananya	#7C4DCC purple	Card, header, buttons
Meera	#2F8F5B green	Card, header, buttons
Rohan	#C8412F red	Card, header, buttons
Kamla	#B8336A magenta	Card, header, buttons

Member page (top to bottom)

- 'Back to All family' link.
- Coloured header: big emoji, name, 'Adult - Mother - ID MQ-003', 'Last updated: not yet' (becomes a date after an edit).
- Two buttons: red **Need help** and outlined **Share update**.
- **Today's medicines**: list, or 'No regular medicines. Add one if needed.' with an 'Add a medicine' field and Add medicine button.
- **Medical update log**: dropdown (starts at 'Doctor visit'), text box 'What changed? e.g. BP 130/80', Save update button, list or 'No updates yet.'
- **Immunization record**: vaccine name, 'Given on' date, 'Next due' date, Add vaccine button, list or 'No vaccines recorded yet.'
- **Profile details**: Date of birth, Blood group, Height, Weight, Allergies, Conditions, Doctor, Emergency contacts, Home care notes (empty ones show 'Not added'), and an Edit details button.

2. What the app does

Feature	Behaviour
Family QR	One QR (poster, fridge, wallet card) opens the home screen above.
Emergency card (no code)	Need help button on every member opens: name, age, blood group, allergies, conditions, emergency contacts as tap-to-call links, and a big 'Call 108' button. Public by design, minimum data only.
Full record (code required)	Medicines, update log, immunisations and all profile fields stay locked until the 6-digit access code is entered. Demo code: **482915**.
Editing	Add medicine, Save update, Add vaccine and Edit details work as in the reference. Changes are saved on the phone (localStorage), stored encrypted with the same key, and shown only after unlock. 'Last updated' changes after each save.
Share update	Uses the phone's share sheet (Web Share API) to send a text summary of the latest update. Falls back to copy-to-clipboard.
Auto-lock	Decrypted data lives in memory only. Locks after 2 minutes idle or when the tab closes.
Mobile number option	Optional passphrase = last 4 digits of the registered mobile + 4-digit PIN. This is only a code format; no SMS is sent in the static demo.

3. Architecture

The site is static. At build time a Node script reads `family.plain.json` (kept on your laptop, never committed). It writes the public emergency data to `emergency.json` and the full records to `data.enc.json`, encrypted.

```
Laptop (private)            GitHub repo (public)             Phone (browser)
-------------------         ------------------------         --------------------
family.plain.json   --build--> docs/emergency.json (plain) -->  Home + Need help card
(not committed)             docs/data.enc.json (AES-GCM) -->  Enter code -> decrypt
                            docs/index.html, app.js, css       in memory (Web Crypto)
                            docs/qr-family.png                 edits -> localStorage (encrypted)
```

Repository structure

```
health-at-home/
|-- docs/                    <- GitHub Pages serves this folder
|   |-- index.html           home screen + member page (single page app)
|   |-- app.js               render, decrypt, edit, share
|   |-- styles.css           dark theme, colour variables per member
|   |-- emergency.json       public minimal data
|   |-- data.enc.json        encrypted full records
|   |-- qr-family.png        generated QR
|   `-- sw.js                optional offline cache
|-- build/
|   |-- family.plain.json    FULL data (in .gitignore, never pushed)
|   |-- encrypt.js           builds emergency.json + data.enc.json
|   `-- make-qr.js           builds the QR image
|-- .gitignore               must contain: build/family.plain.json
|-- package.json             dependency: qrcode
`-- README.md
```

4. Data model (one member)

```
{
  "id": "MQ-005", "name": "Kamla Sharma", "role": "Grandmother", "group": "Elder",
  "color": "#B8336A", "avatar": "elder-woman",
  "emergency": {                        // goes to emergency.json (plain)
    "age": 72, "bloodGroup": "B+",
    "allergies": ["None known"],
    "conditions": ["Hypertension", "Type 2 diabetes"],
    "contacts": [{"name": "Rohan Sharma (Son)", "phone": "+91 90000 00003"},
                 {"name": "Meera Sharma (Daughter-in-law)", "phone": "+91 90000 00004"}]
  },
  "full": {                             // goes to data.enc.json (encrypted)
    "dob": "1954-06-05", "sex": "Female", "heightCm": 155, "weightKg": 62,
    "doctor": "Dr. S. Verma, City General Hospital (fictional)",
    "homeCareNotes": "Fall risk, low vision (sample)",
    "medicines": [{"name": "Metformin", "dose": "500 mg", "when": "twice daily after meals"}],
    "updateLog": [{"date": "2026-10-25", "type": "Doctor visit", "text": "BP 128/80, medicines unchanged"}],
    "immunizations": [{"vaccine": "Influenza", "given": "2026-10-12", "nextDue": "2027-10-12"}],
    "lastUpdated": "2026-10-25"
  }
}
```

5. Encryption method (what the developer implements)

- Build step (Node): random 16-byte salt and 12-byte IV; key = PBKDF2(code, salt, 600000, SHA-256, 256 bit); ciphertext = AES-256-GCM over the JSON of all 'full' blocks.
- Output `data.enc.json`: {salt, iv, iterations, ciphertext} in base64.
- Browser: `crypto.subtle.importKey`, then `deriveKey`, then `decrypt`. If decrypt throws, show 'Incorrect code'.
- Use a fresh random IV every time a blob is encrypted, including each time the phone saves an edit.
- After 5 wrong tries, disable the form for 30 seconds (slows casual guessing only).

6. Fictional family dataset - The Sharma family, Delhi

Ages are as on 9 November 2026. Aarav, Ananya and Kamla match the profile details seen in the reference video. The video showed Meera's and Rohan's fields empty ('Not added'), so their details, and Ananya's doctor, are invented here. Contacts: Rohan +91 90000 00003, Meera +91 90000 00004 (fake numbers).

MQ-001 - Aarav Sharma (Son, Paediatrics)	
Age / DOB / sex	5 years / 12 March 2021 / Male
Blood group	O positive
Height / weight	110 cm / 18 kg
Allergies	**Peanuts: urticaria, lip swelling. No known drug allergy.**
Conditions	Peanut allergy (one hospital admission for an allergic reaction)
Doctor	Dr. A. Mehta, City Children's Hospital (fictional)
Emergency contacts	Meera Sharma (Mother), Rohan Sharma (Father), Kamla Sharma (Grandmother)
Home care notes	Emergency: follow the prescribed allergy action plan.
Today's medicines	No regular medicines. Adrenaline auto-injector as prescribed in the doctor's allergy plan (carried by parents).
Latest vitals	BP 100/62 mmHg (2026-09-14, paediatric check)
Immunization record	DTP booster: given 20 Mar 2026

MQ-002 - Ananya Sharma (Daughter, Adolescent)	
Age / DOB / sex	15 years / 20 August 2011 / Female
Blood group	B positive
Height / weight	158 cm / 48 kg
Allergies	**No food or drug allergy. Triggers: dust, pollen, smoke, strenuous exercise.**
Conditions	Bronchial asthma (diagnosed at 8 years), two past exacerbations, no admissions
Doctor	Dr. P. Nair, City Children's Hospital (fictional)
Emergency contacts	Meera Sharma (Mother), Rohan Sharma (Father)
Home care notes	Carry reliever inhaler to school and sports. Avoid dusty rooms.
Today's medicines	Salbutamol inhaler 100 mcg, 1-2 puffs when needed for wheeze (reliever)
Latest vitals	BP 108/68 mmHg; peak flow in normal range (2026-09-14)
Immunization record	Td/Tdap: given 25 Aug 2021, next due Aug 2031

MQ-003 - Meera Sharma (Mother, Adult)	
Age / DOB / sex	38 years / 14 February 1988 / Female
Blood group	A positive
Height / weight	160 cm / 58 kg
Allergies	**None known**
Conditions	Migraine without aura (since 2015)
Doctor	Dr. S. Rao, City General Hospital (fictional)
Emergency contacts	Rohan Sharma (Husband), Kamla Sharma (Mother-in-law)
Home care notes	Rest in a dark room during an attack. Keep hydrated.
Today's medicines	Paracetamol 500 mg when needed for headache, maximum 4 doses in 24 hours
Latest vitals	BP 112/72 mmHg; haemoglobin 12.6 g/dL (2026-10-25)
Immunization record	Influenza: given 12 Oct 2026, next due Oct 2027

MQ-004 - Rohan Sharma (Father, Adult)	
Age / DOB / sex	42 years / 3 October 1984 / Male
Blood group	B positive
Height / weight	174 cm / 80 kg
Allergies	**Dust (sneezing). No drug allergy.**
Conditions	Dyslipidaemia (since 2023)
Doctor	Dr. R. Kapoor, City General Hospital (fictional)
Emergency contacts	Meera Sharma (Wife), Kamla Sharma (Mother)
Home care notes	Annual lipid profile each October.
Today's medicines	Atorvastatin 10 mg once daily at night
Latest vitals	BP 126/82 mmHg; LDL cholesterol 132 mg/dL (2026-10-25)
Immunization record	Influenza: given 12 Oct 2026, next due Oct 2027

MQ-005 - Kamla Sharma (Grandmother, Elder)	
Age / DOB / sex	72 years / 5 June 1954 / Female
Blood group	B positive
Height / weight	155 cm / 62 kg
Allergies	**None known**
Conditions	Hypertension, type 2 diabetes (sample)
Doctor	Dr. S. Verma, City General Hospital (fictional)
Emergency contacts	Rohan Sharma (Son), Meera Sharma (Daughter-in-law)
Home care notes	Fall risk, low vision (sample)
Today's medicines	Metformin 500 mg twice daily after meals; Amlodipine 5 mg once daily in the morning
Latest vitals	BP 128/80 mmHg; fasting glucose 118 mg/dL (2026-10-25)
Immunization record	Influenza: given 12 Oct 2026, next due Oct 2027; Pneumococcal: given 15 Feb 2024

Consistency check: blood groups work genetically (Rohan B+ and Meera A+ can have O+ and B+ children; Kamla B+ and Rohan B+ are compatible). Ages match dates of birth on 9 Nov 2026. Update-log entries start empty ('No updates yet') as in the reference; the Kamla sample entry in section 4 is optional.

7. Security design and honest limitations

Topic	Design / limitation
Protected	Full records are AES-256-GCM encrypted. Changing the file makes decryption fail.
Public on purpose	Emergency card data (age, blood group, allergies, conditions, contacts). Like a medical ID bracelet, it must work when nobody knows the code.
Offline guessing	The encrypted file is downloadable from a public site, so an attacker can try codes offline. A 6-digit code has only 1,000,000 combinations: fine for a demo, weak for real data. PBKDF2 slows each guess but does not stop it.
Stronger version (say in Q and A)	Use a long passphrase, or put decryption behind a backend (Firebase / Supabase) that rate-limits attempts, sends an OTP to the registered mobile and logs access.
Local edits	Edits saved on the phone are encrypted with the same key. Anyone with the unlocked phone can still read them, so also use the 2-minute auto-lock.
Real data	Health data about real people is covered by India's DPDP Act 2023. A real deployment needs consent, a backend and proper hosting.

8. Step-by-step build and deploy

```
1. Create a GitHub repo named health-at-home. Add the folders from section 3.
2. Put the 5 members from section 6 into build/family.plain.json (format in section 4).
3. Add build/family.plain.json to .gitignore BEFORE the first commit.
4. npm init -y && npm install qrcode
5. node build/encrypt.js 482915       -> writes docs/emergency.json, docs/data.enc.json
6. node build/make-qr.js https://<your-username>.github.io/health-at-home/
                                      -> writes docs/qr-family.png
7. git add . && git commit -m "Health at Home demo" && git push
8. GitHub: Settings -> Pages -> Source: Deploy from branch -> main, folder /docs.
9. Open the Pages URL on a phone, scan docs/qr-family.png to test.
10. Print the QR on A4. Test it from 1 metre away and in low light.
```

9. Acceptance checklist (give this to the developer)

- Home screen matches the reference: teal hero card, 5 coloured member cards with correct ages and group labels.
- Member page has all 5 sections in the reference order, with Need help and Share update buttons.
- Need help opens the emergency card with tap-to-call contacts and Call 108, without any code.
- Code 482915 unlocks the full records; any other code shows 'Incorrect code'.
- Add medicine, Save update, Add vaccine and Edit details work, persist after reload, and update 'Last updated'.
- No plain full-record data in the repo, in the network tab, or in localStorage.
- Looks right at 360 px width (phone). QR scans from paper and the live page loads in under 3 seconds.

10. Demo script for the competition (3 minutes)

Time	What you do / say
0:00	Problem: in an emergency at home, family members cannot recall allergies, medicines or vaccine dates. Paper files get lost.
0:30	Scan the QR with a phone on the projector. Show the 5 colour-coded family cards.
1:00	Open Aarav (5). Tap Need help: peanut allergy and contacts appear instantly, with a Call 108 button.
1:30	Tap unlock, enter a wrong code to show rejection, then 482915. Show Kamla's medicines, vitals and immunisations.
2:00	Add an update ('BP 130/80', Doctor visit) and a vaccine. Show 'Last updated' change and the Share update sheet.
2:20	Show the GitHub repo: only data.enc.json is there, unreadable. One sentence on AES-256-GCM and PBKDF2.
2:40	Limitations and next step: OTP to registered mobile and a rate-limited backend for real use.

Likely judge questions

- **Why is the emergency card not encrypted?** It must work when nobody knows the code, like a medical ID bracelet. It holds only the minimum.
- **Is it truly secure?** It stops casual access. A 6-digit code can be guessed offline on a public static site, so a real product needs a backend with rate limiting and OTP.
- **Why GitHub Pages?** Free, no server cost, and any team can fork and deploy it.
- **Extensions?** Medicine reminders, appointment tracker, multi-family support, Hindi and Bengali toggle.

11. Next step

Share the competition rules and judging criteria so the brief can be adjusted to any mandatory fields. The complete working code (index.html, app.js, styles.css, encrypt.js, make-qr.js) can be written next so the repo is ready to push.